Impact of Adversarial Attacks on Spiking Neural Networks

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INTRODUCTION

Overview: Does spike encoding strategy affect adversarial robustness in SNNs, and do SNNs resist attacks differently from ANNs and CNNs?

WHY THIS MATTERS

- SNNs are increasingly deployed on edge & neuromorphic hardware for safety-critical tasks
- Adversarial robustness of SNNs remains understudied vs. classical deep nets
- Encoding strategy (rate vs. temporal) may introduce implicit robustness but no systematic study exists

SPIKE ENCODING STRATEGIES

Rate Coding

Pixel $x \in [0, 1]$ sets spike **probability** per step:

$s_t \sim \text{Bernoulli}(x),\ t = 1, \ldots, T$

Bright → many spikes. Dense representation.

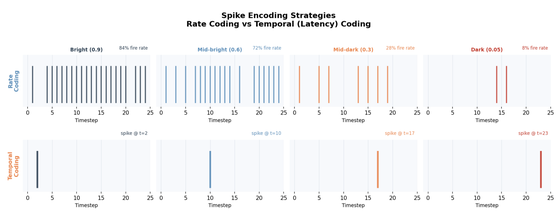
$T = 25$ timesteps (MNIST); $T = 10$ (CIFAR rate).

Temporal Coding

Pixel sets **first-spike latency**:

$t_{\text{spike}} = \lfloor (1 - x)(T - 1) \rfloor$

Bright → fires early. **One spike per neuron.** Extremely sparse. $T = 25$ (MNIST & CIFAR temporal).



*Rate coding uses Bernoulli sampling (MNIST) or deterministic repeat (CIFAR, pixels normalized to $\mathbb{R}$).

ARCHITECTURE

SNN-MNIST (LIF-MLP)

Input (784) → FC(1000) → LIF → FC(10) → LIF

$T = 25$ steps · $\beta = 0.90$ · Adam (2×10^{-3}) · Cosine LR

SPIKING CIFAR (CONV-SNN)

Conv(64)→BN→LIF→AvgPool

Conv(128)→BN→LIF→AvgPool

Conv(256)→BN→LIF→AvgPool

Dropout(0.4) · FC(512)→BN→LIF · FC(10)→LIF

$T_{\text{rate}} = 10,\ \beta = 0.50$ · $T_{\text{temp}} = 25,\ \beta = 0.90$

· Cosine LR (rate $= 1e^{-3}$, temporal $= 5e^{-4}$)

SURROGATE GRADIENT

Heaviside spike function → non-differentiable. We replace its derivative with a fast-sigmoid surrogate $\sigma'(U)$ during backprop :

$$\sigma'(U) = \frac{\text{slope}}{(1 + |\text{slope} \cdot U|)^2}$$

THE SPIKING NEURON MODEL

We use the **Leaky Integrate-and-Fire (LIF)** model (discrete time):

$$U_i[t] = \beta\, U_i[t-1] + \sum_j W_{ij}\, S_j[t] - V_{\text{th}}\, S_i[t-1]$$

Integrates input spikes $S_j[t]$, fires when $U_i[t] > V_{th}$, resets by subtraction, and decays via leakage factor β.

MODELS EVALUATED

Model Architecture	Dataset	Coding
SNN-Rate	MNIST / CIFAR-10	Rate (LIF)
SNN-Temporal	MNIST / CIFAR-10	Latency (LIF)
ANN (MLP)	MNIST	ReLU Baseline
CNN (VGG-style)	CIFAR-10	Standard Baseline

All SNN models use Surrogate Gradient Descent for backpropagation.

TRAINING • Surrogate gradient learning (Neftci et al., 2019) via **snnTorch**. Adam optimiser, cosine annealing LR schedule. 5 runs with different seeds (MNIST); 1 run (CIFAR-10).

ADVERSARIAL ATTACKS

ATTACK PIPELINE

Clean image x → ∇ loss (surrogate) → Perturb $x + \delta$ → Encode → Spikes

Straight-through estimator (STE) used for temporal encoding at attack time: discrete round/argmax is non-differentiable, so the continuous pixel image is repeated T times for gradient flow.

FGSM — **Fast Gradient Sign Method.** Single step: $x_{\text{adv}} = x + \varepsilon \cdot \text{sign}(\nabla_x \mathcal{L})$. Fast worst-case perturbation within ℓ^∞ ball.

PGD — **Projected Gradient Descent.** 10-step iterative attack, random init inside ε-ball, $\alpha = \varepsilon/4$. Strictly stronger than FGSM — upper-bounds robustness.

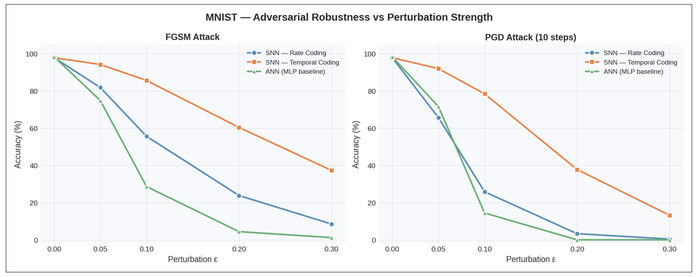
ADVERSARIAL RESULTS: MNIST & CIFAR-10

MNIST
$\varepsilon = 0.20$ • Moderate attack strength
$N_{\text{eval}} = 1000$ • 5-seed mean reported

CIFAR-10
$\varepsilon = 0.05$ • Moderate attack strength
$N_{\text{eval}} = 1000$ • Conv×3 → FC → LIF

Model (MNIST)	Clean (ε=0)	FGSM (ε=0.2)	PGD (ε=0.2)
SNN — Rate Coding	97.9%	23.8%	3.3%
SNN — Temporal Coding	97.8%	60.4%	37.8%
ANN (MLP baseline)	98.4%	4.4%	0.0%

Model (CIFAR-10)	Clean (ε=0)	FGSM (ε=0.05)	PGD (ε=0.05)
SNN — Rate Coding	69.6%	33.4%	15.7%
SNN — Temporal Coding	79.3%	43.4%	28.8%
CNN (VGG-style baseline)	89.5%	24.0%	8.5%



CLEAN VS ADVERSARIAL SAMPLES

Figure 1. **MNIST — Adversarial Samples ($\varepsilon = 0.2$).** Rows: SNN-Rate · SNN-Temporal · ANN baseline. Green border = correct prediction; red = fooled.

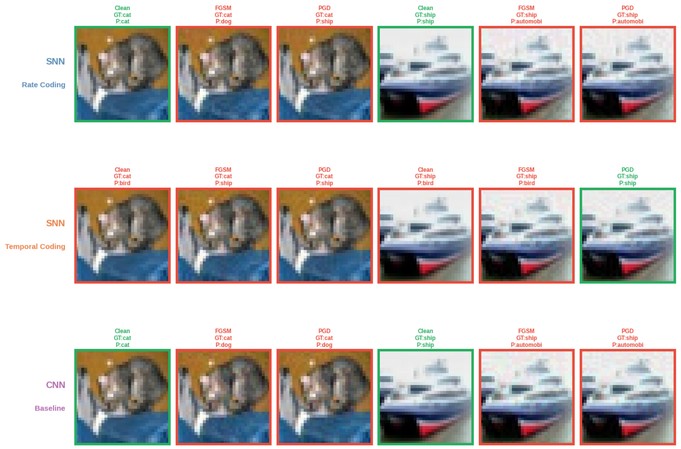
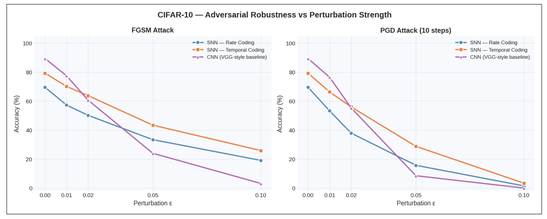


Figure 2. **CIFAR-10 — Adversarial Samples ($\varepsilon = 0.1$).** Rows: SNN-Rate · SNN-Temporal · CNN baseline. Each group: Clean · FGSM · PGD.



KEY FINDINGS

1. **Temporal ≫ Rate under attack.** MNIST: temporal retains 60.4% (FGSM) vs. Rate's 23.8% at $\varepsilon = 0.2$ — a **+36.6 pp** advantage. Under PGD: 37.8% vs. 3.3% (**+34.5 pp**).
2. **Both SNNs crush ANNs adversarially.** ANN collapses to 0.0% under PGD $\varepsilon = 0.2$. SNN-Temporal retains 37.8%. Even SNN-Rate: 3.3%.
3. **PGD far stronger than FGSM.** SNN-Rate drops an extra **20.5 pp** from FGSM to PGD at $\varepsilon = 0.2$. ANN already collapsed.
4. **Temporal wins on CIFAR-10 too.** Despite lower clean accuracy (79.3% vs. CNN's 89.5%), temporal SNN retains 28.8% under PGD vs. CNN's 8.5% (**+20.3 pp**).
5. **Encoding = robustness design parameter.** Same architecture, same training — only encoder changes, yet robustness gap is dramatic. Spike timing discreteness impedes gradient-based attacks.

FUTURE WORK

- **Adversarial training** via surrogate-gradient PGD for certified robust neuromorphic models.
- Evaluate on **larger architectures** (ResNet-SNN) and neuromorphic datasets (DVS-Gesture, N-MNIST).
- Investigate **black-box & transfer attacks** where gradient sparsity advantage may differ.
- Explore **hybrid / population coding** robustness profiles.
- Robustness analysis on actual **neuromorphic hardware** (Intel Loihi, BrainScaleS).

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